

FX6m
高硬钢
FX6m Hardened
Steels

碳素钢
Carbon Steel

合金钢
Alloy Steel

调质钢
Prehardened Steel

高硬度钢
Hardened Steel

高硬度钢
Hardened Steel

高硬度钢
Hardened Steel

微小径
Micro Diameter

高硬钢用2刃微小径平底铣刀
Hardened Steels 2-Flute Micro Diameter Square Endmills

2刃·平底铣刀
2-Flute·Square

平面
Planing

侧切
Side Cutting

切槽
Slotting

粗加工
Roughing

半精加工
Semi Finishing

精加工
Finishing

切削条件
Cutting Conditions

外径公差
Disturbance

D≤0.1: 0~-0.005
0.15≤D≤0.9: 0~-0.007

h5
0~-0.005

(mm)

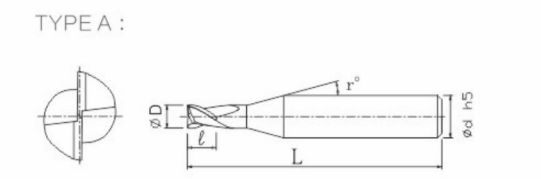
铜合金
Copper

- 在硬度领域 (48~65HRC) 的长时间加工中也可实现稳定的长寿命。
- 改进了刀刃形状，改善了倾斜量和加工精度。
- Realized stably long tool life against high hardened steels (48 ~ 65HRC).
- Improved milling deflection and accuracy by advanced flute design.

产品代码 Code No.	规格型号 Spec Typ.	外径(D) Dia.	刃长(l) Length of Cut	颈角(r) Neck Taper Angle	柄径(d) Shank Dia.	全长(L) Overall Length	形状 Type	定价 Retail Price
FX6MSE00032	0.03x0.045L	0.03	0.045	11°	4	45	A	
FX6MSE00042	0.04x0.06L	0.04	0.06	11°	4	45	A	
FX6MSE00052	0.05x0.075L	0.05	0.075	11°	4	45	A	
FX6MSE00062	0.06x0.09L	0.06	0.09	11°	4	45	A	
FX6MSE00072	0.07x0.105L	0.07	0.105	11°	4	45	A	
FX6MSE00082	0.08x0.12L	0.08	0.12	11°	4	45	A	
FX6MSE00092	0.09x0.135L	0.09	0.135	11°	4	45	A	
FX6MSE0012	0.1x0.15L	0.1	0.15	15°	4	45	A	
FX6MSE00152	0.15x0.2L	0.15	0.2	15°	4	45	A	
FX6MSE0022	0.2x0.4L	0.2	0.4	15°	4	45	A	
FX6MSE00252	0.25x0.5L	0.25	0.5	15°	4	45	A	
FX6MSE0032	0.3x0.6L	0.3	0.6	15°	4	45	A	
FX6MSE00352	0.35x0.7L	0.35	0.7	15°	4	45	A	
FX6MSE0042	0.4x0.8L	0.4	0.8	15°	4	45	A	
FX6MSE00452	0.45x0.9L	0.45	0.9	15°	4	45	A	
FX6MSE0052	0.5x1L	0.5	1	15°	4	45	A	
FX6MSE0062	0.6x1.2L	0.6	1.2	15°	4	45	A	
FX6MSE0072	0.7x1.4L	0.7	1.4	15°	4	45	A	
FX6MSE0082	0.8x1.6L	0.8	1.6	15°	4	45	A	
FX6MSE0092	0.9x1.8L	0.9	1.8	15°	4	45	A	

平底
Square

高硬度材料加工用。超群的切削排出性能，可实现高效率加工。
For machining of high-hardness materials. Excellent chip removal performance enables high efficiency.



硬质合金
X6m
涂层
65
HRC
切削条件
2

Micro Diameter
微小径

切削参数参考表
Recommended Milling Conditions

FX6m
高硬钢
FX6m Hardened
Steels

碳素钢
Carbon Steel

合金钢
Alloy Steel

调质钢
Prehardened Steel

高硬度钢
Hardened Steel

高硬度钢
Hardened Steel

高硬度钢
Hardened Steel

加工材料 Work Material	高速钢 High Speed Tool Steels SKH (~65HRC)				淬火钢 Hardened Steels SKD11 · PD613 (~62HRC)				淬火钢 Hardened Steels SKD61 · STAVAX · HPM-38 (~52HRC)				碳素钢·调质钢 Carbon Steels · Prehardened Steels S50C · NAK55 · NAK80 · HPM-1 (~43HRC)			
	主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
外径 Dia.																
0.03	40,000	10	0.001	0.005	40,000	10	0.001	0.01	40,000	20	0.001	0.01	40,000	10	0.001	0.01
0.04	40,000	20	0.001	0.008	40,000	20	0.001	0.01	40,000	30	0.001	0.01	40,000	10	0.001	0.01
0.05	40,000	30	0.001	0.008	40,000	20	0.001	0.01	40,000	40	0.001	0.02	40,000	10	0.002	0.02
0.06	40,000	40	0.001	0.01	40,000	30	0.001	0.02	40,000	50	0.002	0.03	40,000	20	0.002	0.03
0.07	40,000	50	0.001	0.01	40,000	40	0.001	0.02	40,000	60	0.002	0.03	40,000	30	0.003	0.03
0.08	40,000	50	0.002	0.02	40,000	60	0.001	0.03	40,000	80	0.002	0.04	40,000	50	0.003	0.04
0.09	40,000	60	0.002	0.02	40,000	80	0.002	0.03	40,000	100	0.003	0.04	40,000	100	0.004	0.05
0.1	40,000	70	0.002	0.03	40,000	100	0.002	0.04	40,000	120	0.003	0.05	40,000	150	0.005	0.06
0.15	40,000	80	0.002	0.04	40,000	100	0.002	0.06	40,000	120	0.003	0.07	40,000	150	0.005	0.09
0.2	30,000	120	0.003	0.06	30,000	160	0.003	0.08	30,000	200	0.003	0.1	30,000	240	0.005	0.12
0.25	30,000	160	0.003	0.07	30,000	200	0.003	0.1	30,000	200	0.003	0.13	30,000	300	0.006	0.14
0.3	30,000	200	0.003	0.09	30,000	250	0.003	0.12	30,000	300	0.003	0.15	30,000	350	0.007	0.18
0.35	28,000	250	0.004	0.1	30,000	300	0.04	0.14	30,000	350	0.004	0.18	30,000	400	0.08	0.2
0.4	25,000	300	0.005	0.12	30,000	350	0.005	0.16	30,000	400	0.005	0.2	30,000	450	0.01	0.24
0.45	23,000	350	0.005	0.14	26,000	400	0.006	0.18	28,000	450	0.008	0.23	30,000	500	0.015	0.28
0.5	20,000	400	0.005	0.15	23,000	450	0.007	0.2	25,000	500	0.01	0.25	30,000	550	0.02	0.3
0.6	20,000	350	0.005	0.18	23,000	400	0.007	0.25	25,000	500	0.01	0.3	30,000	550	0.02	0.35
0.7	20,000	320	0.007	0.21	23,000	400	0.015	0.28	25,000	560	0.02	0.35	25,000	690	0.03	0.4
0.8	20,000	550	0.012	0.24	23,000	650	0.03	0.32	25,000	780	0.04	0.4	25,000	850	0.05	0.45
0.9	20,000	600	0.014	0.28	23,000	700	0.04	0.4	25,000	850	0.05	0.5	23,000	900	0.06	0.5
备注 Notes	※本切削参数仅供参考。请根据实际的加工形状及使用机床等调整切削参数。 ※切深量的ap表示轴向切入量，ae表示步距量。 ※加工淬火钢时，建议使用油雾冷却方式。 ※轴向进刀建议采用螺旋进刀及倾斜进刀方式。 ※沟槽切削时建议参考切削参数表，切深量ap及进给速度设定为50%以下，采用来回切削加工方式。 ※发生振刀时，请以相同的比率降低主轴转速和进给速度。此外，主轴转速过低时，也以相同的比率调。 ※These recommended cutting conditions indicate just reference. It should be adjusted according to milling shape and machine type. ※ap:Axial Depth of Cut, ae:Radial Depth of Cut. ※Recommend to apply herical or ramping for approaching into axial direction. ※recommend To Apply Herical Or Ramping For Approaching Into Axial Direction. ※For slotting, recommend reciprocating milling by adjusting feed & ap in below 50% of recommended milling condition. ※Reduce both spindle speed and feed at same rate for chattering and also for insucient spindle speed of a machine.															

平底
Square